Maulik Madhavi

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https://github.com/maulikmadhavi

Experience

NCS Pte Ltd, Singapore, Senior AI Engineer — Apr 2021 – present, 5 years 7 months

- Contributed to KAI Create and KAI SnS, building and optimizing computer-vision and backend components for low-code video analytics and search/summarization workflows.
- Designed and implemented a million-scale face indexing system and person/vehicle detection pipelines with attribute recognition.
- Built a prompt studio for efficient video search, summarization, and analysis, working across product, engineering, and research teams.
- Fine-tuned open-weight LLMs (Mistral-7B, Llama-2-7B) and VLMs (Qwen3-VL, Moondream) with LoRA/SFT on internal Alpaca-style instruction datasets.
- Optimized vLLM serving for high-throughput video description generation, producing summaries for 48 concurrent 10-second chunks in ~7 seconds on a single H100 GPU.
- Optimized computer-vision models for edge and on-prem deployment through quantization, pruning, and distillation using ONNX Runtime, TensorRT, and OpenVINO.
- Developed an agentic AI workflow for autonomous computer-vision model training across planning, data collection, annotation, training, and evaluation.

National University of Singapore (NUS), Singapore, Research Fellow — Dec 2017 – Apr 2021, 3 years 5 months

- Designed and deployed a backend dialogue system for autonomous vehicles supporting multimodal conversations (FAQ, intent-driven, and open-domain) using Google Search API and Wikipedia.
- Developed a text-dependent speaker verification system optimized for mobile deployment.
- Improved ASR for healthcare by adapting language models to Singaporean English, yielding significant WER reductions.
- Built an automated form-filling system for SCDF call centers using NLU to extract actionable information from ASR transcripts.
- Mentored and co-supervised undergraduate and graduate students on research projects.

Dhirubhai Ambani Institute of Information and Communication Technology (DA-IICT), Gandhinagar, Project Personnel — Apr 2012 – June 2014, 2 years 3 months

- Prosodically guided phonetic engine development for Gujarati and Marathi languages (Advisor: Prof. Hemant Patil)
- Developed a phonetic engine for IPA transcription, syllabification, and prosodic analysis across read, spontaneous, and lecture speech from multiple dialectal regions.
- Designed the end-to-end phonetic engine workflow using HMM-based phoneme recognition, spectral analysis for syllable segmentation, and source-characteristic analysis for suprasegmental and acoustic modeling.
- Built an audio search pipeline leveraging the phonetic engine for spoken-content retrieval and discovery in Indian language processing.
- Explored query-by-audio alignment techniques to improve representation and matching for spoken-content retrieval, laying the foundation for Ph.D. research.

Education

Ph.D. **Dhirubhai Ambani Institute of Information and Communication Technology (DA-IICT), Gandhinagar**, Information and Communication Technology (ICT) — July 2011 – Nov 2017
- Ph.D. in Information and Communication Technology (ICT)
- CPI: 9.75/10
- Thesis: Design of QbE-STD System: Audio Representation and Matching Perspective

M.Tech **Dhirubhai Ambani Institute of Information and Communication Technology (DA-IICT), Gandhinagar**, Information and Communication Technology (ICT) — July 2009 – June 2011
- Master of Technology (M.Tech) in Information and Communication Technology (ICT)
- Communication Systems specialization
- CPI: 8.26/10
- Thesis: Person Recognition from Their Hum

B.E. **V.V.P. Engineering College, Saurashtra University, Rajkot**, Electronics and Communication (EC) — July 2005 – June 2009
- Bachelor of Engineering (B.E.) in Electronics and Communication (EC)
- Percentage: 69.73% (aggregate)

Publications

Vocal Tract Length Normalization using a Gaussian mixture model framework for query-by-example spoken term detection — Nov 2019
M. C. Madhavi, H. A. Patil
10.1016/j.csl.2019.03.005 (Computer, Speech and Language, Elsevier, vol. 58, pp. 175-202)

Design of mixture of GMMs for query-by-example spoken term detection — Nov 2018
M. C. Madhavi, H. A. Patil
10.1016/j.csl.2018.04.006 (Computer, Speech and Language, Elsevier, vol. 52, pp. 41-55)

Combining evidences from magnitude and phase information using VTEO for person recognition using humming — Nov 2018
H. A. Patil, M. C. Madhavi
10.1016/j.csl.2018.04.006 (Computer Speech and Language, Elsevier, vol. 52, pp. 225-256)

Partial matching and search space reduction for QbE-STD — Sep 2017
M. C. Madhavi, H. A. Patil
10.1016/j.csl.2018.04.006 (Computer, Speech & Language, Elsevier, vol. 45, pp. 58-82)

Static and dynamic information derived from source and system features for person recognition from humming — Sep 2012
H. A. Patil, M. C. Madhavi, K. K. Parhi
10.1007/s10772-012-9161-5 (Int. Journal of Speech Technology, Springer, vol. 15, no. 3, pp. 393-406)

Spoken Keyword Retrieval using Source and System Features — Dec 2017
M. C. Madhavi, H. A. Patil, N. Bhendawade
10.1007/978-3-319-69900-4_42 (PReMI, Lecture Notes in Computer Science (LNCS), pp. 333-341)

VTLN Using different warping functions for template matching Mar 2016
M. C. Madhavi, S. Sharma, H. A. Patil
10.1007/978-3-319-30315-4_10 (Machine Intelligence and Big Data in Industry, Springer International Publishing, pp. 111-121)

Vocal tract length normalization features for audio search Sep 2015
M. C. Madhavi, S. Sharma, H. A. Patil
10.1007/978-3-319-24033-6_44 (Int. Conf. Text, Speech, and Dialogue, TSD, pp. 387-395)

Combining evidences from temporal and spectral features for person recognition from humming Feb 2012
H. A. Patil, M. C. Madhavi, R. Jain, A. Jain
10.1007/978-3-642-27387-2_40 (PerMIn, Lecture Notes in Computer Science (LNCS), vol. 7143, pp. 321-328, Springer-Verlag)

Speaker recognition using sparse representation via superimposed features Dec 2013
Y. Gaur, M. C. Madhavi, H. A. Patil
10.1007/978-3-642-45062-4_19 (PReMI, Lecture Notes in Computer Science (LNCS), vol. 8251, pp. 140-147, Springer-Verlag)

Exploring teacher-student learning approach for multi-lingual speech-to-intent classification Dec 2021
B. Sharma, M. Madhavi, X. Zhou, H. Li
10.1109/ASRU51503.2021.9688041 (IEEE Automatic Speech Recognition and Understanding Workshop (ASRU), pp. 419-426)

Knowledge Distillation from BERT Transformer to Speech Transformer for Intent Classification Aug-Sep 2021
Y. Jiang, B. Sharma, M. Madhavi, H. Li
10.21437/Interspeech.2021-402 (Interspeech, Brno, Czech Republic, pp. 4713-4717)

Diagnosis of COVID-19 using Auditory Acoustic Cues Aug-Sep 2021
R. Das, M. Madhavi, H. Li
10.21437/Interspeech.2021-497 (Interspeech, Brno, Czech Republic, pp. 921-925)

Multi-target DoA estimation with an audio-visual fusion mechanism Jun 2021
X. Qian, M. Madhavi, Z. Pan, J. Wang, H. Li
10.1109/ICASSP39728.2021.9413776 (IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Toronto, Canada, pp. 4280-4284)

Leveraging Acoustic and Linguistic Embeddings from Pretrained Speech and Language Models for Intent Classification Jun 2021
B. Sharma, M. Madhavi, H. Li
10.1109/ICASSP46034.2021.9413388 (IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Toronto, Canada, pp. 7498-7502)

OpenNLU: Open-Source Web-Interface NLU Toolkit for Development of Conversational Agent Dec 2020
Y. Ong, M. Madhavi, K. Chan
10.1109/APSIPA48774.2020.9306398 (Asia-Pacific Signal and Information Processing Association (APSIPA) Annual Summit and Conference (ASC), Auckland, New Zealand, pp. 381-385)

Query-by-Example Spoken Term Detection using Generative Adversarial Network Dec 2020
N. Shah, Sreeraj R, M. Madhavi, N. Shah, H. Patil
10.1109/APSIPA48774.2020.9306256 (APSIPA Annual Summit and Conference (ASC), Auckland, New Zealand, pp. 644-648)

Transformer-based Arabic Dialect Identification Dec 2020
W. Lin, M. Madhavi, R. Das, H. Li
10.1109/IALP51396.2020.9310504 (International Conference on Asian Language Processing (IALP), Kuala Lumpur, Malaysia, pp. 192-196)

Speaker-Utterance Dual Attention for Speaker and Utterance Verification Oct 2020
T. Liu, R. Das, M. Madhavi, S. Shen, H. Li
10.21437/Interspeech.2020-2656 (Interspeech, Shanghai, China, pp. 4293-4297)

RSL2019: A Realistic Speech Localization Corpus Oct 2019
R. Sheelvant, B. Sharma, M. Madhavi, R. Das, S.R.M. Prasanna, H. Li
10.1109/O-COCOSDA46657.2019.9060842 (Oriental COCOSDA, Cebu City, Philippines, pp. 1-6)

A Unified Framework for Speaker and Utterance Verification Sept 2019
T. Liu, M. Madhavi, R. Das, H. Li
10.21437/Interspeech.2019-1994 (Interspeech, Graz, Austria, pp. 4320-4324)

First Leap Towards Development of Dialogue System for Autonomous Bus Apr 2019
M. Madhavi, T. Zhan, H. Li, M. Yuan
10.1007/978-981-15-9323-9_37 (International Workshop on Spoken Dialogue Systems Technology (IWSDS), Sicily, Italy, pp. 1-6)

Compensating Utterance Information in Fixed Phrase Speaker Verification Nov 2018
R. Das, M. C. Madhavi, H. Li
10.1109/APSIPA.2018.8659742 (APSIPA Annual Summit and Conference (ASC), Honolulu, Hawaii, USA, pp. 1708-1712)

Replay Spoof Detection using Power Function Based Features Nov 2018
P. A. Tapkir, M. R. Kamble, H. A. Patil, M. C. Madhavi
10.1109/APSIPA.2018.8659582 (APSIPA Annual Summit and Conference (ASC), Honolulu, Hawaii, USA, pp. 1019-1023)

Unsupervised Vocal Tract Length Warped Posterior Features for Non-Parallel Voice Conversion Sept 2018
N. Shah, M. C. Madhavi, H. Patil
(Interspeech, Hyderabad, India, pp. 1968-1972)

VTLN-Warped Gaussian posteriorgram for QbE-STD Aug-Sept 2017
M. C. Madhavi, H. A. Patil
10.23919/EUSIPCO.2017.8081270 (25th European Signal Processing Conference (EUSIPCO), Kos Island, Greece, pp. 593-597)

Two Stage Zero-resource Approaches for QbE-STD Dec 2017
M. C. Madhavi, H. A. Patil
10.1109/ICAPR.2017.8593033 (International Conference on Advances in Pattern Recognition (ICAPR), Bangalore, India)

Combining Evidences from Detection Sources for Query-by-Example Spoken Term Detection Dec 2017
M. C. Madhavi, H. A. Patil
10.1109/APSIPA.2017.8282106 (APSIPA Annual Summit and Conference, Kuala Lumpur, Malaysia)

Native language identification using spectral and source-based features Sept 2016
A. Rajpal, T. B. Patel, H. B. Sailor, M. C. Madhavi, Hemant A. Patil, Hiroya Fujisaki
10.21437/Interspeech.2016-1100 (Interspeech, San Francisco, USA, pp. 369-372)

Modification in sequential dynamic time warping for fast computation of query-by-example spoken term detection task June 2016
M. C. Madhavi, H. A. Patil
10.1109/SPCOM.2016.7746657 (International Conference on Signal Processing and Communications (SPCOM), Bangalore, India, pp. 1-6)

Spectral transition measure for detection of obstruents Aug-Sept 2015
M. C. Madhavi, H. A. Patil, B. B. Vachhani
10.1109/EUSIPCO.2015.7362399 (23rd European Signal Processing Conference (EUSIPCO), Nice, France, pp. 330-334)

Significance of phase-based features for person recognition using humming Feb 2015
H. B. Sailor, M. C. Madhavi, H. A. Patil
10.1145/2708463.2709035 (2nd International Conference on Perception and Machine Intelligence (PerMin), Kolkata, pp. 99-103)

A spectral transition measure based MELCEPSTRAL features for obstruent detection Oct 2014
B. Vachhani, K. D. Malde, M. C. Madhavi, H. A. Patil
10.1109/IALP.2014.6973511 (International Conference on Asian Language Processing (IALP), Kuching, Malaysia, pp. 50-53)

Vocal Tract Length Normalization for Vowel Recognition in Low Resource Languages Oct 2014
S. Sharma, M. C. Madhavi, H. A. Patil
10.1109/IALP.2014.6973516 (International Conference on Asian Language Processing (IALP), Kuching, Malaysia, pp. 54-57)

Development of language resources for speech application in Gujarati and Marathi Oct 2014
M. C. Madhavi, S. Sharma, H. A. Patil
10.1109/IALP.2014.6973517 (International Conference on Asian Language Processing (IALP), Kuching, Malaysia, pp. 115-118)

Exploiting speech source information for vowel landmark detection for low resource language Sept 2014
A. Undhad, H. Patil, M. C. Madhavi
10.1109/ISCSLP.2014.6936660 (9th International Symposium on Chinese Spoken Language Processing (ISCSLP), Singapore, pp. 546-550)

Deterministic annealing EM algorithm for developing TTS System in Gujarati Sept 2014
N. Shah, H. Patil, M. Madhavi, H. Sailor, T. Patel
10.1109/ISCSLP.2014.6936666 (9th International Symposium on Chinese Spoken Language Processing (ISCSLP), Singapore, pp. 526-530)

Exploiting variable length Teager energy operator in melcepstral features for person recognition from humming Sept 2014
M. Madhavi, H. A. Patil
10.1109/ISCSLP.2014.6936654 (9th International Symposium on Chinese Spoken Language Processing (ISCSLP), Singapore, pp. 624-628)

Development of vocal tract length normalized phonetic engine for Gujarati and Marathi languages Sept 2014
S. Sharma, M. C. Madhavi, H. A. Patil
10.1109/COCOSDA.2014.7051439 (17th Oriental COCOSDA, Phuket, Thailand)

Development of speech corpora in Gujarati and Marathi for phonetic transcription Nov 2013
K. D. Malde, B. B. Vachhani, M. C. Madhavi, N. H. Chhayani, H. A. Patil
10.1109/COCOSDA.2013.6709865 (International Conference Oriental COCOSDA, Gurgaon, India, pp. 1-6)

Phonetic transcription of Fricatives and Plosives for Gujarati and Marathi Languages Nov 2012
H. A. Patil, M. C. Madhavi, K. D. Malde, B. B. Vachhani
10.1109/IALP.2012.6473725 (International Conference on Asian Language Processing (IALP), Hanoi, Vietnam, pp. 177-180)

Person Recognition Using Humming, Singing and Speech Nov 2012
H. A. Patil, M. C. Madhavi, N. H. Chhayani
10.1109/IALP.2012.6473718 (International Conference on Asian Language Processing (IALP), Hanoi, Vietnam, pp. 149-152)

Significance of magnitude and phase information via VTEO for humming based biometrics Mar-Apr 2012
H. A. Patil, M. C. Madhavi
10.1109/ICB.2012.6199779 (International Conference on Biometrics (ICB), New Delhi, India, pp. 372-377)

Combining evidence from spectral and source-like features for person recognition from humming Aug 27-31 2011
H. A. Patil, M. C. Madhavi, K. K. Parhi
10.21437/Interspeech.2011-132 (Interspeech, Florence, Italy, pp. 369-372)

Awards

IAPR Travel Scholarship: International Association for Pattern Recognition Travel Scholarship for presenting a paper at the International Conference on Biometrics (ICB 2012), Delhi, India. Paper: Significance of Magnitude and Phase Information via VTEO for Humming-Based Biometrics. (Grant reference: IAPR009)

GATE 2009 Qualification: Qualified in Electronics and Communication Engineering with All India Rank 2142

Invited Talks

Doctoral Consortium Talk: Design of QbE-STD System: Audio Representation and Matching Perspective at Summer School on Speech Signal Processing, DA-IICT, Gandhinagar

Overview of Query-by-Example Spoken Term Detection: Talk at ISCA supported Summer School on Advances in Speech and Audio Processing (ASAP), DA-IICT, Gandhinagar

Query-by-Example Spoken Term Detection - A template matching approach: 5-minute Ph.D. competition at the ISCA-supported Summer School on Speech Signal Processing (S4P), DA-IICT, Gandhinagar

Conferences and Workshops

International Conferences Attended: APSIPA-ASC 2020 (Auckland, New Zealand), IALP 2020 (Kuala Lumpur, Malaysia), INTERSPEECH 2019 (Graz, Austria), INTERSPEECH 2018 (Hyderabad, India), IALP 2014 (Kuching, Malaysia), Oriental COCOSDA 2013 (Gurgaon, India), ICB 2012 (New Delhi, India)

Workshops Attended: IWSDS 2019 (Sicily, Italy), IWSDS 2018 (Singapore), ASAP 2016 (DA-IICT, Gandhinagar), S4P 2016 (DA-IICT, Gandhinagar), WiSSAP 2015 (DA-IICT, Gandhinagar), CEP Workshop 2014 (DA-IICT, Gandhinagar)

Professional Activities

Organizing Committee: Publication chair, Web content Manager, IEEE Automatic Speech Recognition and Understanding Workshop, 14-18 Dec 2019, Singapore

Program Committee: Program committee member and Session chair for IEEE International Conference on Asian Language Processing (IALP) 2020

Conference Reviewer: INTERSPEECH (2018-2026), ICASSP (2021, 2023-2026), APSIPA 2018, IALP (2019-2023, 2025-26), ISCSLP 2024, Computer Speech and Language, Speech Communication (Elsevier), IEEE Access

Volunteer: IWSDS 2018 (Singapore), S4P 2016 (DA-IICT), WiSSAP 2015 (DA-IICT), CEP Workshop 2014 (DA-IICT)

Skills

Research Areas: Speech signal processing, voice search, speaker verification, spoken language understanding, deep learning for spoken language processing, million-scale indexing

Programming Languages: Bash, C/C++, Python, JavaScript, MATLAB

Deep Learning & ML Frameworks: TensorFlow, PyTorch, OpenCV, scikit-learn, Hugging Face Transformers, NumPy, Pandas, Matplotlib

Large Language Models & AI Tools: LangChain, LlamaIndex, DSPy, vLLM, RASA, n8n, Agentic Harness (Claude Code, OpenCode, Copilot, make_my_harness)

LLM/VLM Fine-tuning & Serving: LoRA/SFT fine-tuning (Mistral-7B, Llama-2-7B), vision-language model tuning (Gemma, Moondream, BLIP, CLIP), high-throughput vLLM serving (PagedAttention, continuous batching, KV cache)

Model Optimization & Edge Deployment: Quantization (INT8/INT4), pruning, knowledge distillation, ONNX Runtime, TensorRT, OpenVINO, model export (ONNX/TorchScript)

Web & API Frameworks: Flask, FastAPI, Streamlit, Gradio

Audio/Speech Tools & Systems: Kaldi, wakeup-word detection, voice activity detection

DevOps & Deployment: Git, Docker, Kubernetes

Operating Systems: Windows and Linux